RES POLYTECHNIC, BANGALORE-91

CIE Practice Test II - Model Answers

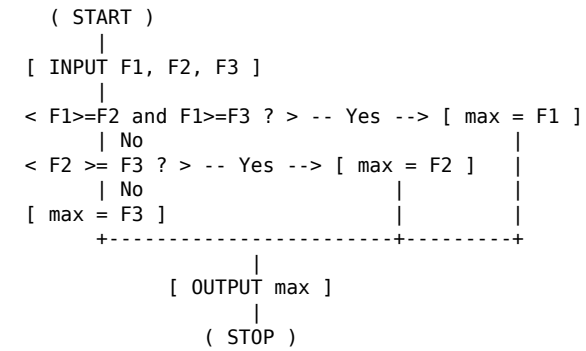
Programming Concepts in Mechanical Engineering | Course Code 25ME34I | Marks 50 | Semester III

Answer any one full question (Q1 or Q2). Both are answered below.

Q1

a) Algorithm and flowchart – maximum of F1, F2, F3 (5 marks)

Algorithm: 1. Start 2. Read F1, F2, F3 3. If F1 ≥ F2 and F1 ≥ F3, max = F1 4. Else if F2 ≥ F3, max = F2 5. Else max = F3 6. Print max 7. Stop



b) Gear system output speed and torque (5 marks)

```
n_in = float(input("Input speed (rpm): "))
T_in = float(input("Input torque (N-m): "))
ratio = float(input("Gear ratio: "))

n_out = n_in / ratio        # speed decreases
T_out = T_in * ratio        # torque increases (ideal, no losses)

print("Output speed =", n_out, "rpm")
print("Output torque =", T_out, "N-m")
```

c) Factorial using while loop (10 marks)

```
n = int(input("Enter a number: "))
fact = 1
i = 1
while i <= n:
    fact = fact * i
    i += 1
print("Factorial of", n, "=", fact)
```

Output (n = 5): `Factorial of 5 = 120`

d) Student list – add, remove, sort (15 marks)

```
students = ["Ravi", "Anu", "Kiran", "Meena"]
print("Original list :", students)

students.append("Suresh")          # add
print("After adding  :", students)

students.remove("Kiran")           # remove
print("After removing:", students)

students.sort()                    # sort
print("After sorting :", students)
```

Output:

```
Original list : ['Ravi', 'Anu', 'Kiran', 'Meena']
After adding  : ['Ravi', 'Anu', 'Kiran', 'Meena', 'Suresh']
After removing: ['Ravi', 'Anu', 'Meena', 'Suresh']
After sorting : ['Anu', 'Meena', 'Ravi', 'Suresh']
```

e) Bank account class (15 marks)

```
class BankAccount:
    def __init__(self, name, balance=0):
        self.name = name
        self.balance = balance
```

```
    def deposit(self, amount):
        self.balance += amount
        print("Deposited:", amount)

    def withdraw(self, amount):
        if amount > self.balance:
            print("Insufficient balance")
        else:
            self.balance -= amount
            print("Withdrawn:", amount)

    def check_balance(self):
        print("Balance =", self.balance)

acc = BankAccount("Ravi", 1000)
acc.deposit(500)
acc.withdraw(2000)      # Insufficient balance
acc.withdraw(300)
acc.check_balance()     # Balance = 1200
```

OR

Q2

a) Algorithm and flowchart – multiplication table of 5 (5 marks)

Algorithm: 1. Start 2. Set i = 1 3. If i ≤ 10, go to step 4, else go to step 6 4. Print 5 × i = 5*i 5. i = i + 1, go to step 3 6. Stop

```
    ( START )
        |
    [ i = 1 ]
        |
   < i <= 10 ? > --- No ---> ( STOP )
        | Yes
   [ PRINT 5 x i = 5*i ]
   [ i = i + 1 ]
        |
        +---------> back to decision
```

b) Arithmetic operations (5 marks)

```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))

print("Addition       =", a + b)
print("Subtraction    =", a - b)
print("Multiplication =", a * b)
if b != 0:
    print("Division       =", a / b)
else:
    print("Division not possible (divide by zero)")
```

c) GCD using a function (10 marks)

```
def gcd(a, b):
    while b != 0:
        a, b = b, a % b
    return a

x = int(input("Enter first number: "))
y = int(input("Enter second number: "))
print("GCD of", x, "and", y, "is", gcd(x, y))
```

Output (48, 36): `GCD of 48 and 36 is 12`

d) Read CSV and count rows (15 marks)

```
import csv

try:
    with open("data.csv", "r") as f:
        reader = csv.reader(f)
        count = 0
        for row in reader:
            count += 1
    print("Number of rows =", count)
except FileNotFoundError:
    print("Error: File not found")
```

`try` runs the file code; if the file is missing, `except` handles the error and the program does not crash.

e) Car class (15 marks)

```
class Car:
    def __init__(self, model, year, color):
        self.model = model
        self.year = year
        self.color = color

    def start(self):
        print(self.model, "started")

    def stop(self):
        print(self.model, "stopped")

    def display(self):
        print("Model:", self.model)
        print("Year :", self.year)
        print("Color:", self.color)

c1 = Car("Swift", 2024, "Red")
c1.display()
c1.start()
c1.stop()
```